



MSc Data Science · Colloquium

Conversational Explanations using Large Language Models on top of the eXplainable AI Question Bank

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Research goal

Can LLMs effectively communicate an AI system's architecture, results, and explanations to domain experts — and how do we measure it?



Why does explainability matter?



Black boxes

Justification for decisions are often non-existent or hidden behind proprietary algorithms.



High-stakes use

Healthcare and other high-risk domains require transparency before AI can be trusted and adopted.



Regulation

The EU AI Act and the GDPR “right to explanation” now require explainability for high-risk systems.



How do we achieve explainability?

Keeping humans in control, responsible, and accountable while opening the black box



Human-centred AI (HC-AI)



Explainable AI (XAI)

Three families of XAI techniques



Intrinsic interpretability



Post-hoc methods



Counterfactuals



LLMs as conversational explanatory agents



The opportunity

- Turn complex topics into accessible narratives.
- Personalised, individualised learning.
- Refine the user's mental models.
- Iteratively fill in knowledge gaps (grounding).
- Ask follow-up questions.



The gap

Prior LLM-for-XAI work is **restricted to templates**, limiting conversational capacity.

Or it is judged only on **user satisfaction & acceptance** — not on whether explanations are correct.

No consensus exists on what to measure, or how.



Thesis contributions

A transferable framework for evaluating LLM-generated explanations



ACF composite metric



Information units



Readability benchmarking



XAI-QB as schema



Instructional template



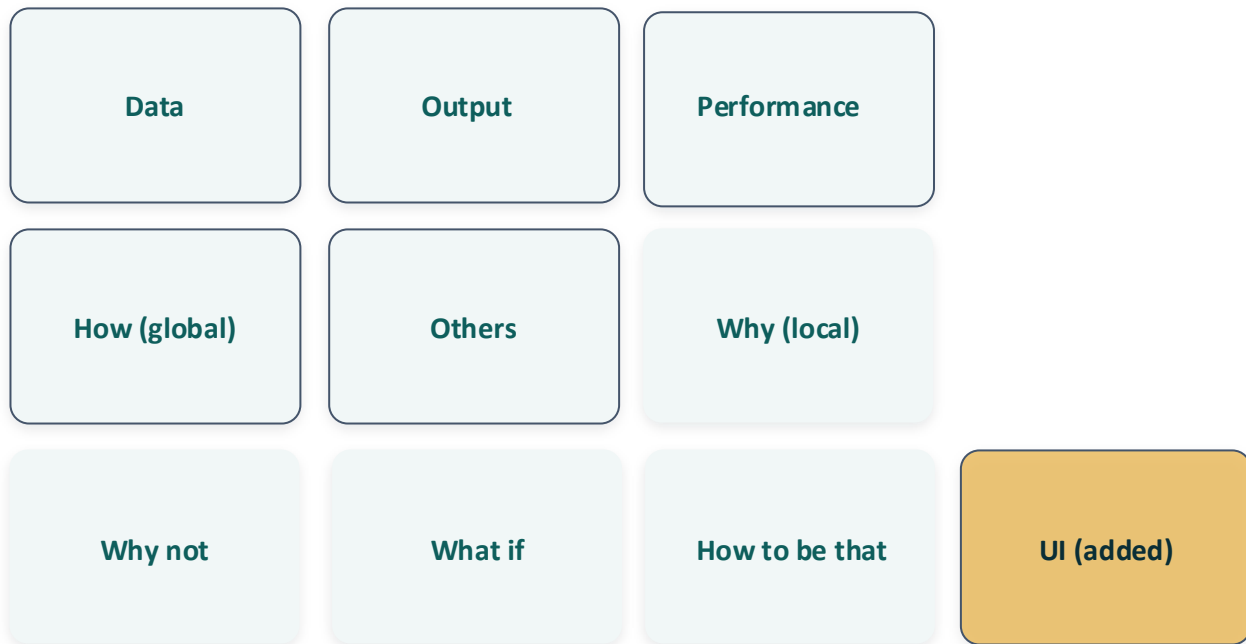
Fine-tuning feedback loops



The eXplainable AI Question Bank (XAI-QB)

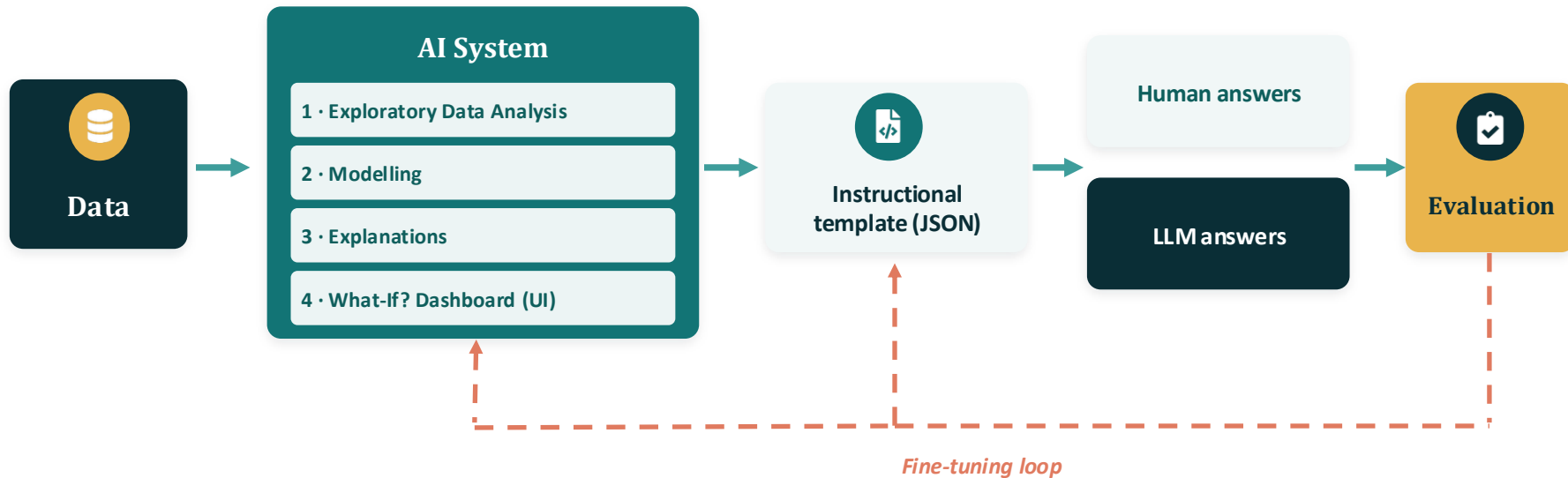
Prototypical questions a user might ask about an AI system

9 explainability-needs categories





Study design

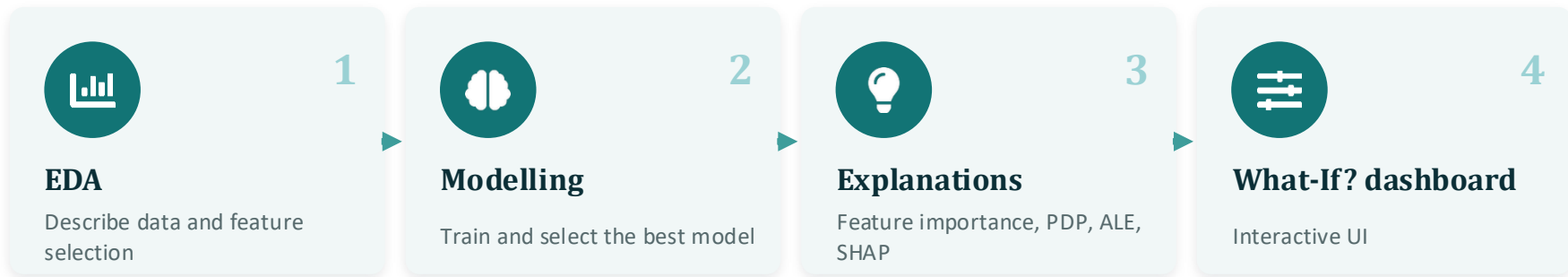




The AI system

A prediction tool and explanation interface for the healthcare domain

Task: Binary classification of H1N1 vaccine uptake *simulating AI-assisted clinical decision-making.*



Data — DrivenData “Flu Shot Learning”, from the US National 2009 H1N1 Flu Survey

Containing protected, demographic, behavioural, health and opinion related features



Modelling



Logistic Regression

Linear baseline



Random Forest

Bagged decision trees

SELECTED



XGBoost

Boosted decision trees



Multi-layer Perceptron

Neural network



Hyperparameter tuning

- Randomised 5-fold cross-validated grid search
- 50 iterations $\times$ 5 folds = 250 fits per model
- Best parameters chosen on ROC-AUC



Explanations — the XAI artefacts

Global, post-hoc methods the LLMs must read and explain



Confusion matrix



AUC-ROC



Feature importance



PDP & ALE



SHAP values



Top features → UI



The What-If? dashboard

An interactive UI – the system’s user-facing component



User controls

- Adjust the 6 most important features via select boxes
- Remaining features imputed to an “average” baseline
- Threshold slider sets how conservative the prediction is
- Outputs: Prediction breakdown, SHAP, PDP plots

Six adjustable features and threshold slider

Doctor recommends H1N1 vaccine

Opinion: H1N1 vaccine is effective

Opinion: H1N1 is a risk

Age group

Race

Sex

0



1



The instructional template

A JSON prompt for structured, repeatable LLM output



Metadata

Title, authors, description, version control.



Persona

“XAI specialist communicating with healthcare professionals.”



Instructions

Guidelines enforcing clear, and clinically relevant answers.
Output and formatting rules.



Questions

The XAI-QB questions, each with an empty string to fill.

Both reference human answers and LLM generated answers were written in the same JSON format.



Experimental setup

Source code and instructional template uploaded to each model's native platform

The three models



ChatGPT 5.2



Claude Opus 4.6



Gemini 3 Pro

3

LLMs evaluated

36

XAI-QB questions

3

runs per model

324

answers generated



Evaluation framework

Three quantitative components combine into one ACF score



Accuracy

Manual scoring vs. human ground-truth

[0, 0.5, 1]



Completeness

Information units found in the answer

$\frac{1}{2}(\text{ROUGE-L} + \text{BERTScore})$



Faithfulness

Semantic similarity to human answer

SBERT

$$\text{ACF} = \frac{1}{3} [\text{Accuracy} + \frac{1}{2}(\text{ROUGE-L} + \text{BERTScore}) + \text{SBERT}]$$

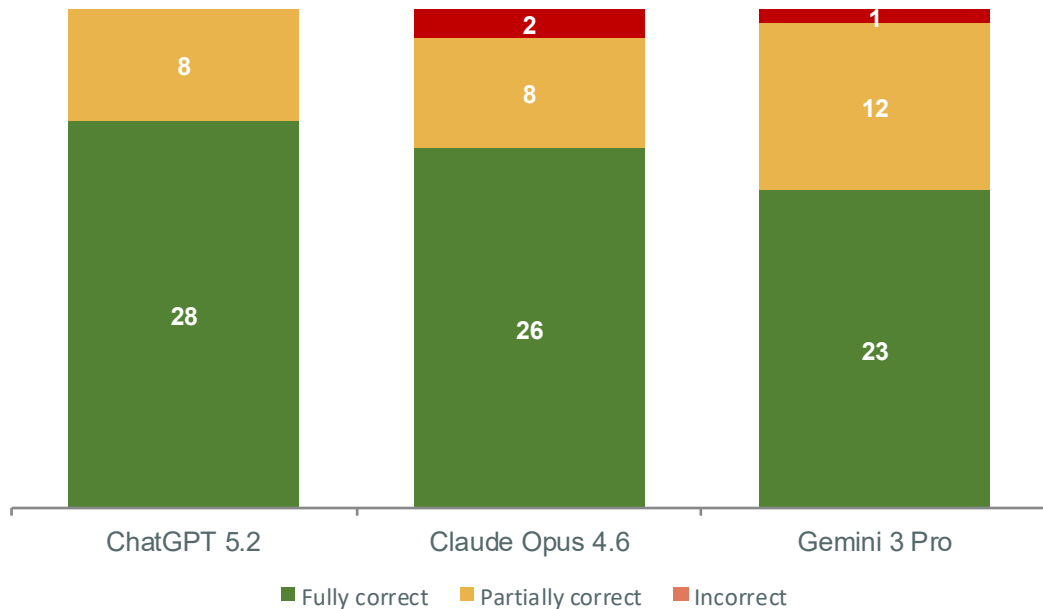
Information units "What is the sample size?" → ["21,365", "37"]

+ 4th dimension: Readability



Results: Accuracy

97% of answers were at least partially correct



Overall accuracy

ChatGPT 5.2 **88.9%**

Claude Opus 4.6 **83.3%**

Gemini 3 Pro **80.6%**

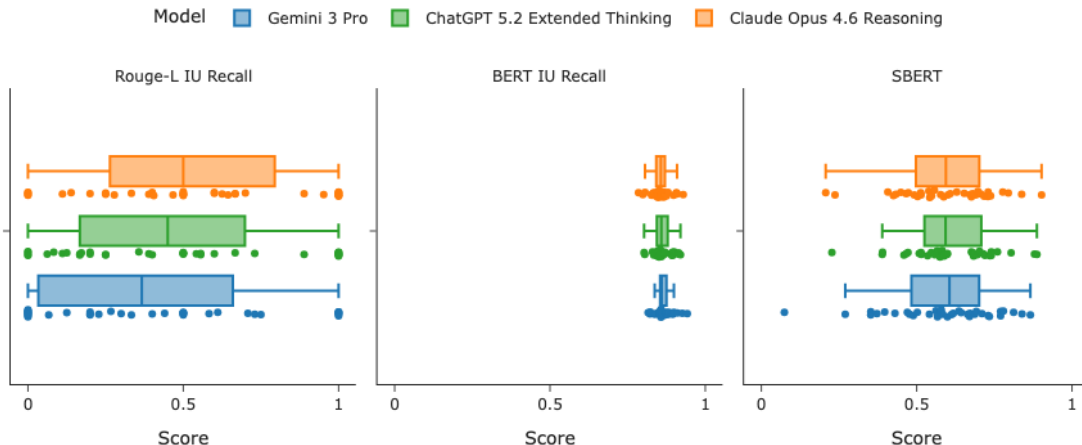
Only 3 incorrect answers
all from misreading, none from hallucination



Results: Completeness and Faithfulness

Comparable performance, with each model strong on a different axis

Overall Similarity Metrics between Human and Large Language Model XAIQB Answers



Components of ACF score

ChatGPT 5.2

Highest SBert. Most faithful answers

Claude Opus 4.6

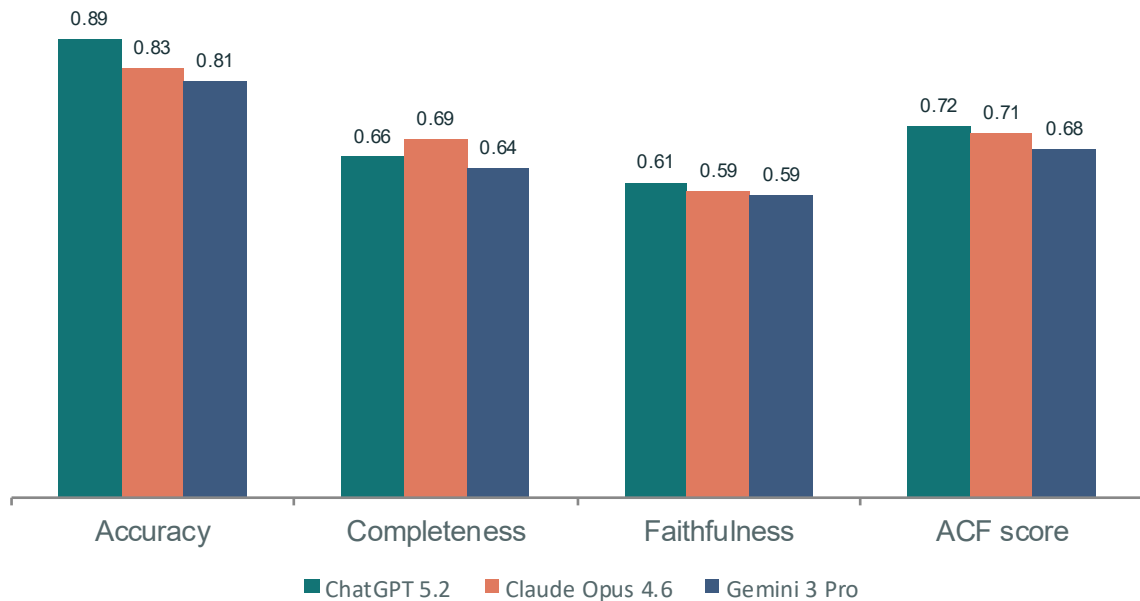
Highest Rouge-L. Most complete answers

Gemini 3 Pro

Highest BertScore. Most concise answers



Results: The ACF score



ACF Score

ChatGPT 5.2

Best overall, highest accuracy and faithfulness

Claude Opus 4.6

Most complete answers

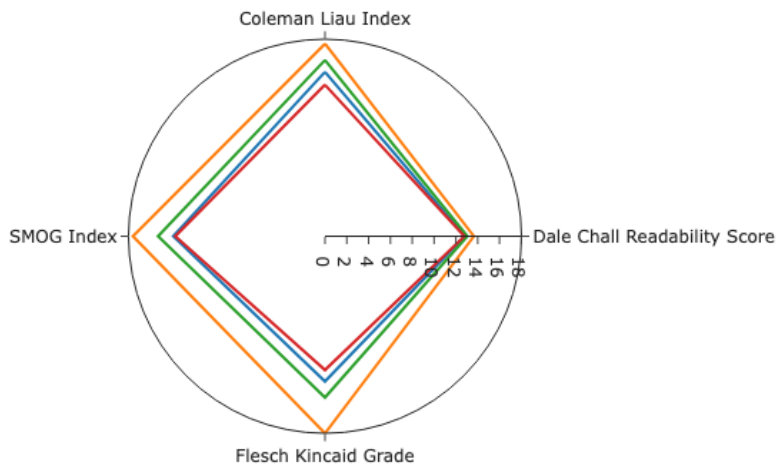
Gemini 3 Pro

Most readable answers



Results: Readability

Model — Gemini 3 Pro — ChatGPT 5.2 Extended Thinking — Claude Opus 4.6 Reasoning — Human



US grade level (lower = easier)

Human

Baseline

13.1

Gemini 3 Pro

Easiest to read LLM

13.8

ChatGPT 5.2

Most consistent

14.8

Claude Opus 4.6

Hardest to read LLM

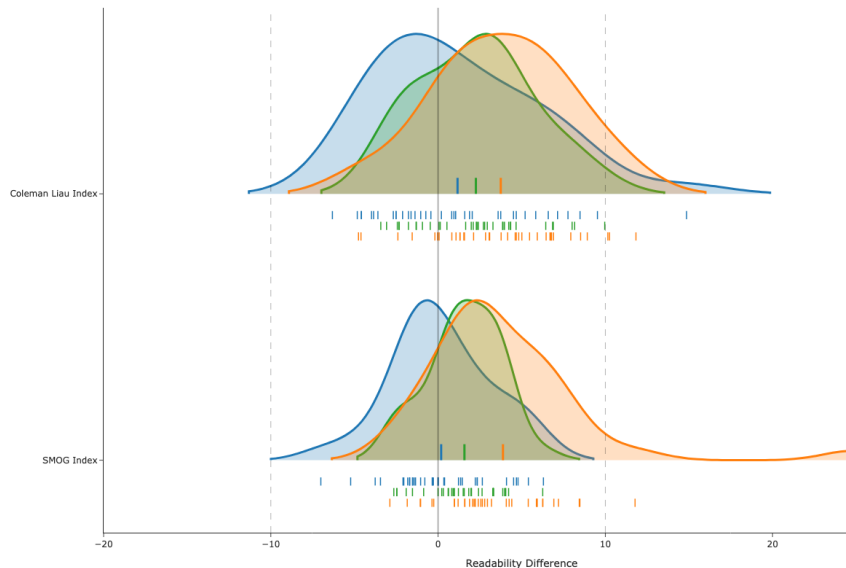
16.7



Results: Readability

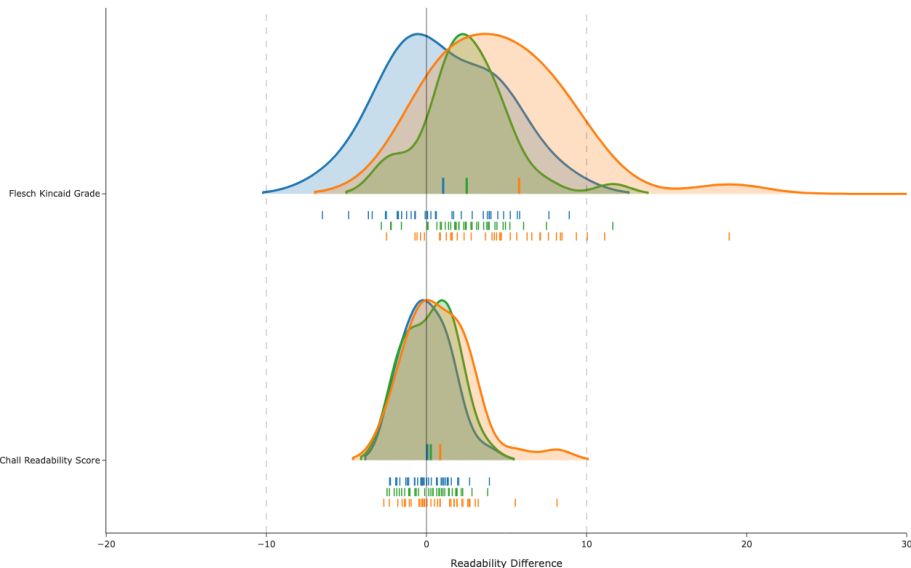
LLM Readability Difference to Human Answers

Model ■ Gemini 3 Pro ■ ChatGPT 5.2 Extended Thinking ■ Claude Opus 4.6 Reasoning



LLM Readability Difference to Human Answers

Model ■ Gemini 3 Pro ■ ChatGPT 5.2 Extended Thinking ■ Claude Opus 4.6 Reasoning



$$\text{Difference } (\Delta) = \text{Readability}_{\text{LLM}} - \text{Readability}_{\text{Human}}$$



Model benchmarking

Different strengths suggest different use-cases

1

ChatGPT 5.2

"Best all-rounder"

- Highest accuracy and faithfulness
- Most consistent readability
- Best overall ACF (0.720)

High-stakes domains where balance matters

2

Claude Opus 4.6

"The specialist"

- Most complete answers
- In-depth, analytical, technical
- Most verbose and hardest to read

Technical audiences who want depth

3

Gemini 3 Pro

"The communicator"

- Most concise and easiest to read
- Trade-off with lowest accuracy

Quick synopses and summaries



Limitations and future work



The fluency trap. A readable but inaccurate answer can feel plausible. This is dangerous when clinicians lack the technical knowledge to verify it. ACF must be paired with readability, plus human-in-the-loop checks.

Limitations

- Single domain, one AI system, one dataset
- One set of (subjective) reference human answers
- Only global questions
- Restricted API access. Reproducibility is empirical

Future work

- Multiple reference answers; more domains and datasets
- Penalise verbose answers in the ACF score
- Local and fairness excluded
- User studies with clinicians (trust, satisfaction)
- LLM-as-a-judge to expedite accuracy evaluation



Conclusion

LLMs can serve as conversational explanatory agents — and the ACF framework makes their explanations measurable.

0.843

Accuracy

0.663

Completeness

0.596

Faithfulness

0.700

Overall ACF

- ✓ Strong performance despite subjective ground-truth and open-ended, technical questions.
- ✓ Distinct model strengths. Suggesting matching the model to the audience and their use-case.
- ✓ ACF also flags vague XAI-QB questions and incoherent AI-system design.



Thank you

Questions & Discussion

Henry El-Jawhari

Freie Universität Berlin · ZKI-PH, Robert Koch Institute



Backup – Full results

Component metrics and verbosity across the three models

ACF components (selected run per model)

Model	Accuracy	ROUGE-L	BERTScore	SBERT	ACF
ChatGPT 5.2	0.889	0.458	0.862	0.611	0.720
Claude Opus 4.6	0.833	0.530	0.857	0.593	0.706
Gemini 3 Pro	0.806	0.407	0.868	0.586	0.676

Verbosity (average per answer)

Metric	ChatGPT 5.2	Claude Opus 4.6	Gemini 3 Pro	Human
Word count	63.7	91.2	37.2	60.8
Words / sentence	22.0	29.3	19.6	21.4
Difficult words	15.0	24.4	7.2	13.2



Backup – ACF by Category

Category	Average ACF score			Average
	ChatGPT _{5.2}	Claude _{Opus 4.6}	Gemini _{3 Pro}	
Data	0.718	0.778	0.781	0.759
Output	0.686	0.644	0.563	0.631
Performance	0.739	0.691	0.674	0.702
How (global model-wide explanations)	0.701	0.702	0.667	0.690
Others	0.773	0.796	0.734	0.767
UI	0.787	0.631	0.817	0.745

Category	Question	Rank	Average ACF score
Data	What is the source of the training data?	1	0.909
How (global)	What kind of algorithm is used?	2	0.889
	...	...	...
How (global)	What is the system's overall logic?	35	0.466
How (global)	How does it weigh different features?	36	0.441



Backup – Readability metrics

Four grade-level formulas, structural and lexical complexity

Flesch–Kincaid Grade

Weights sentence length and syllables per word.

SMOG Index

Based on polysyllabic words; common for health literacy.

Coleman–Liau Index

Uses characters, not syllables. Easy to compute.

Dale–Chall Score

Penalises words outside a list of 3,000 familiar words.

All map to a US grade level, easily compared to the education level of the target audience.



Backup – Instructional Template

```
{
  "metadata": {
    "template_name": "Healthcare XAI Question Bank Instructional Template",
    "version": "1.3",
    "last_updated": "2025-01-25",
    "authors": "Henry El-Jawhari, Robert Koch Institute (RKI), Freie Universität
↳ Berlin",
    "description": "An instructional JSON template for a Large Language Model to
↳ generate structured, clinically relevant explanations about an AI system for a
↳ healthcare audience, based on the eXplainable AI (XAI) Question Bank."
  },
  "persona": {
    "persona": "You are an Explainable AI specialist communicating
↳ professionals about the safe, transparent, and ethical use
↳ in clinical and public health settings.",
    "target_audience": "Healthcare professionals (e.g. clinicians,
↳ researchers)",
    "tone": "Maintain a professional, trust-building tone suitable
↳ communication. It should be factual, clear, and safety-conscious."
```

```
"questions": {
  "Data": {
    "What kind of data was the system trained on?": {"answer": ""},
    "What is the source of the training data?": {"answer": ""},
    "How were the labels/ground-truth produced?": {"answer": ""},
    "What is the sample size of the training data?": {"answer": ""},
    "What dataset(s) is the system NOT using?": {"answer": ""},
```



Backup – Information Units

Defining core terms which need to be included in an answer

Q: *“What is the sample size of the training data?”*

Human Answer

There are a total of 26,707 observations with 37 features. This was split into training and testing data using an 80:20 split. Training data is 21,365 rows, testing data is 5,342 rows.



Information units: [“21,365”, “37”]

As validation, information units need to have a ROUGE-L ≥ 0.8 when measuring against its reference human answer.



Backup – Highest ACF

Category	Data
Question	What is the source of the training data?
Model	Claude _{Opus 4.6}

ROUGE-L Recall	Bert Recall	SBERT	Accuracy	ACF score
0.952	0.909	0.902	1	0.944

Information units: [“DrivenData”, “Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines”, “US National 2009 H1N1 Flu Survey (NHFS)”]

Human Answer: “Data is sourced from DrivenData’s ‘Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines’ competition and originates from the US National 2009 H1N1 Flu Survey (NHFS). This survey was conducted between October 2009 and June 2010 as a one-time telephone survey of households to monitor vaccinations during the 2009-2010 flu season in response to the 2009 H1N1 pandemic.”

LLM Answer: “The training data originates from the National 2009 H1N1 Flu Survey (NHFS), which was sponsored by the National Center for Immunization and Respiratory Diseases (NCIRD) and conducted jointly by NCIRD and the National Center for Health Statistics (NCHS) at the Centers for Disease Control and Prevention (CDC). The NHFS was a list-assisted random-digit-dialling telephone survey of households in the United States, conducted between October 2009 and June 2010. The dataset was accessed via DrivenData’s public competition ‘Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines’”



Backup – Lowest ACF

Category	How (global model-wide explanation)
Question	How does it weigh different features?
Model	Claude _{Opus 4.6}

ROUGE-L Recall	Bert Recall	SBERT	Accuracy	ACF score
0	0.834	0.207	0.5	0.375

Information Units: [“Split nodes”, “Reduce error (loss function)”, “Gain”]

Human Answer: “XGBoost splits nodes by maximising gain, which measures how much a split reduces the prediction error (loss function). Each feature’s usefulness is measured by how often and how effectively it is chosen to split nodes during tree construction. Features that produce greater reductions in loss get selected more often. Internally, each split’s ‘gain’ (improvement in the objective) is stored - this can be viewed as a weight for that feature.”

LLM Answer: “Variable importance analysis reveals that the system assigns the highest importance to: opinion on H1N1 vaccine effectiveness, doctor recommendation for H1N1 vaccine, and opinion on H1N1 risk. These opinion and recommendation features have substantially higher importance scores than other variables. Features grouped as ‘opinion’ and ‘doctor recommendation’ categories collectively have the greatest impact on model performance. Behavioural features (e.g. hand washing, mask use) and protected demographic features (e.g. age, sex, race) have comparatively lower individual importance, though they still contribute to the overall prediction. Geographic and household variables have the least influence.”